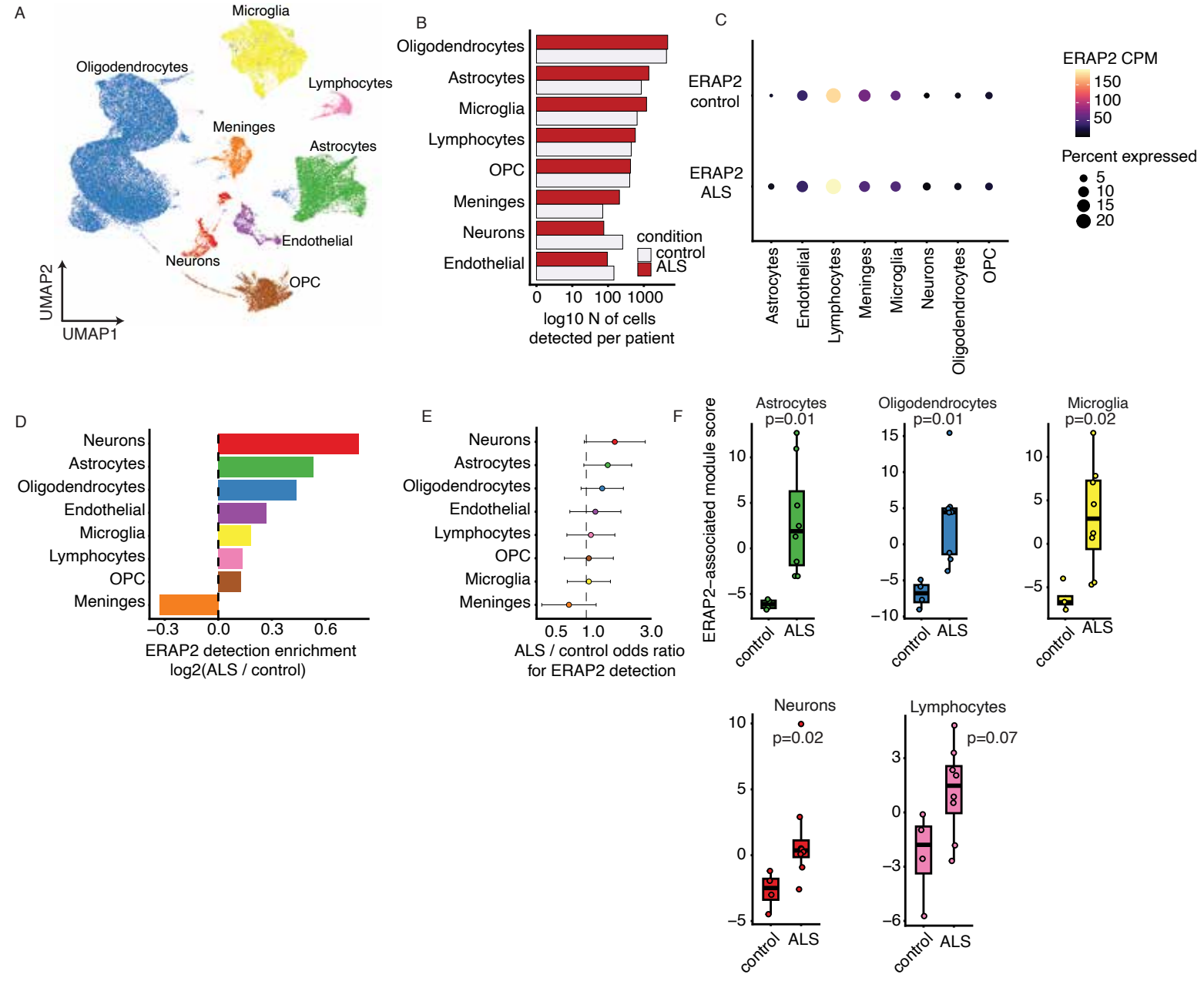

A
Microglia
Oligodendrocytes
Lymphocytes
Meninges
Astrocytes
Endothelial
Neurons
OPC
UMAP2
UMAP1
B
Oligodendrocytes
Astrocytes
Microglia
Lymphocytes
OPC
Meninges
Neurons
Endothelial
condition
control
ALS
0 10 100 1000
log10 N of cells
detected per patient
C
ERAP2
control
ERAP2
ALS
Astrocytes
Endothelial
Lymphocytes
Meninges
Microglia
Neurons
Oligodendrocytes
OPC
ERAP2 CPM
150
100
50
Percent expressed
5
10
15
20
D
Neurons
Astrocytes
Oligodendrocytes
Endothelial
Microglia
Lymphocytes
OPC
Meninges
−0.3 0.0 0.3 0.6
ERAP2 detection enrichment
log2(ALS / control)
E
Neurons
Astrocytes
Oligodendrocytes
Endothelial
Lymphocytes
OPC
Microglia
Meninges
0.5 1.0 3.0
ALS / control odds ratio
for ERAP2 detection
F
ERAP2−associated module score
Astrocytes
p=0.01
Oligodendrocytes
p=0.01
Microglia
p=0.02
Neurons
p=0.02
Lymphocytes
p=0.07
control
ALS